SOLAR DEWATERING

DRIVING SUSTAINABLE MINING FORWARD

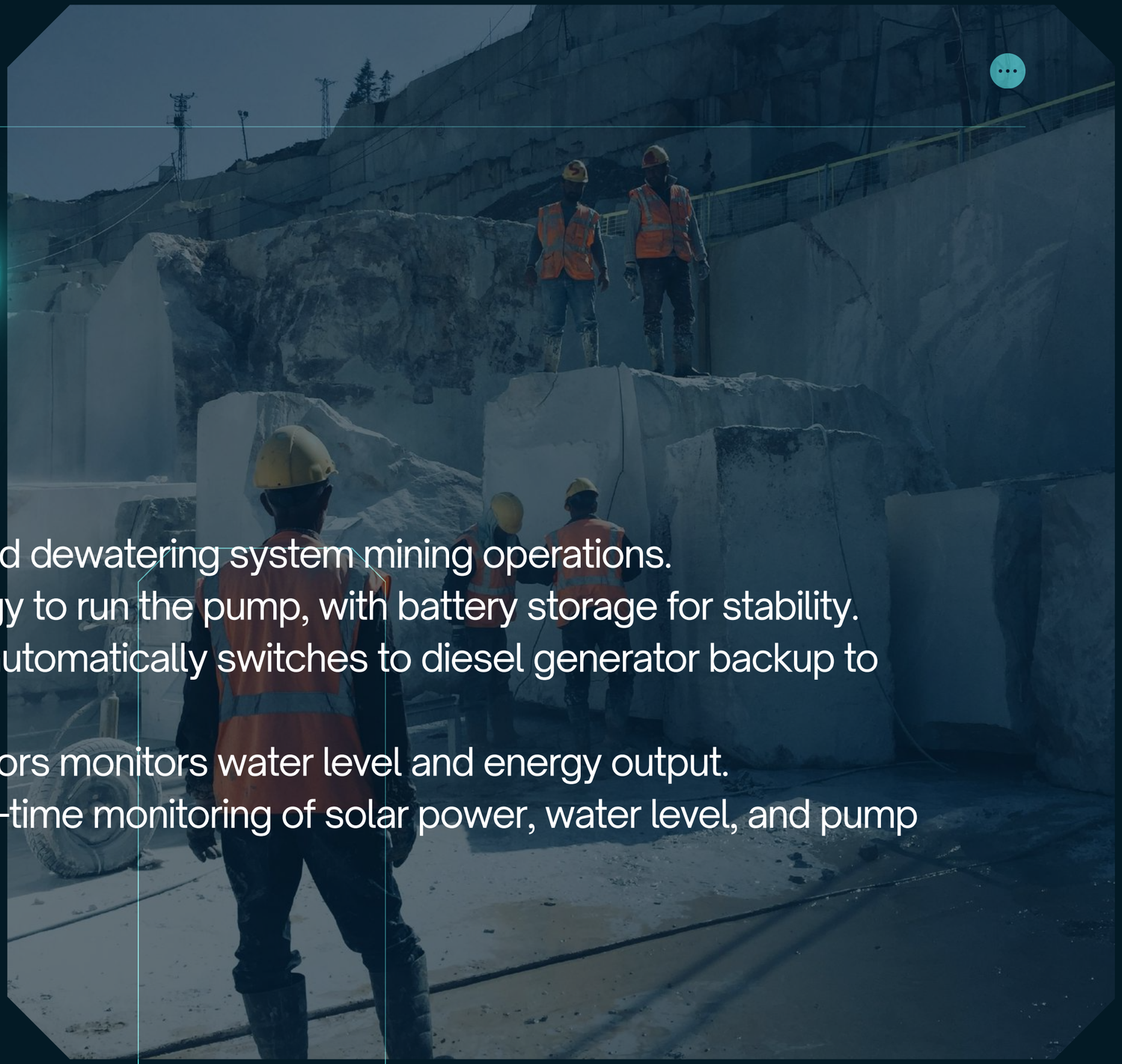
powered with

Terravolts

PROBLEM STATEMENT

- Copper mines continuously accumulate groundwater and rainwater, requiring constant dewatering for safe operations.
- Conventional systems rely heavily on diesel or grid-powered pumps, resulting in high operational costs and fuel dependency.
- Remote mining locations make fuel transportation and maintenance difficult.
- Diesel usage also increases carbon emissions and environmental impact.
- Hence, there is a need for a smart, sustainable, and energy- efficient dewatering solution.

PROPOSED SOLUTION



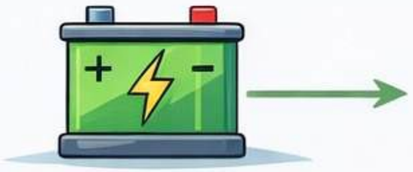
- we proposes a solar powered automated dewatering system mining operations.
- solar photovoltaic panels generate energy to run the pump, with battery storage for stability.
- if solar power is insufficient, the system automatically switches to diesel generator backup to ensure continuous operation.
- an ESP32 controller with relays and sensors monitors water level and energy output.
- a mobile app and lcd display provide real-time monitoring of solar power, water level, and pump status.

PROJECT COMPONENTS

WHY WE USE THEM

Solar Panel

Generates renewable electricity for the system



Battery

Stores solar energy for continuous operation

Potentiometer

Used for calibration/ testing thresholds

Voltage Sensor

Monitors solar and battery voltage

Water Flow Sensor

Measures pumped water flow rate

Water Level Sensor

Detects water level in the mine pit

DC Pump

Removes water from the mine

ESP32 Controller

Controls system and processes sensor data

LCD Display

Shows real-time system information

Buzzer & LED

Gives warning alerts and system status

Buck Converter

Converts voltage to stable 5V supply

Relay Module

Automatically switches the pump

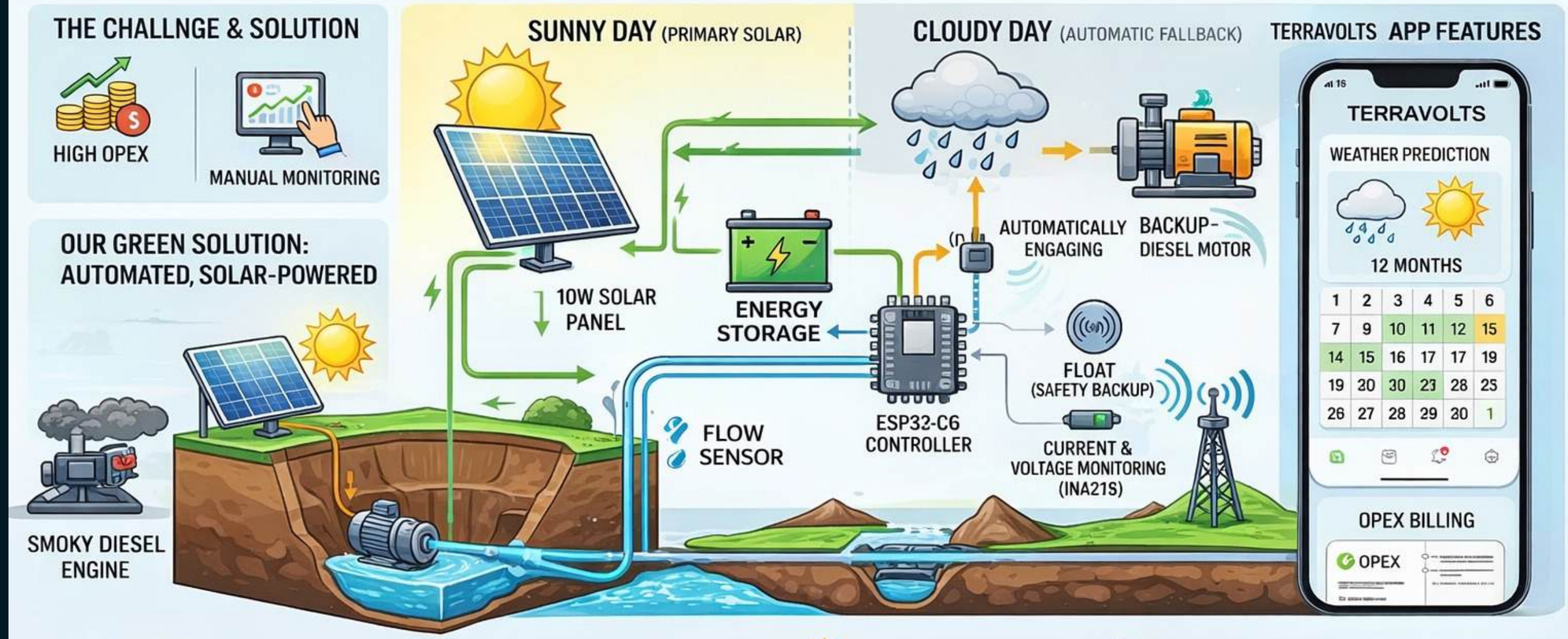
Servo + LDR

Adjusts solar panel toward sunlight

TERRAVOLTS DYNAMICS

SOLAR SMART MINE DEWATERING SYSTEM

Renewable Energy & Sustainable Development



LOWER OPERATIONAL COSTS (OPEX)

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ZERO-EMISSION PRIORITY

AUTOMATED MINE SAFETY

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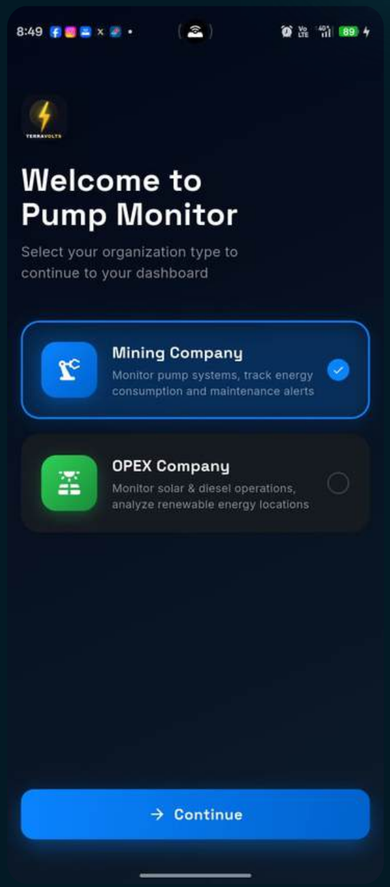
OUR VERY OWN APP!!

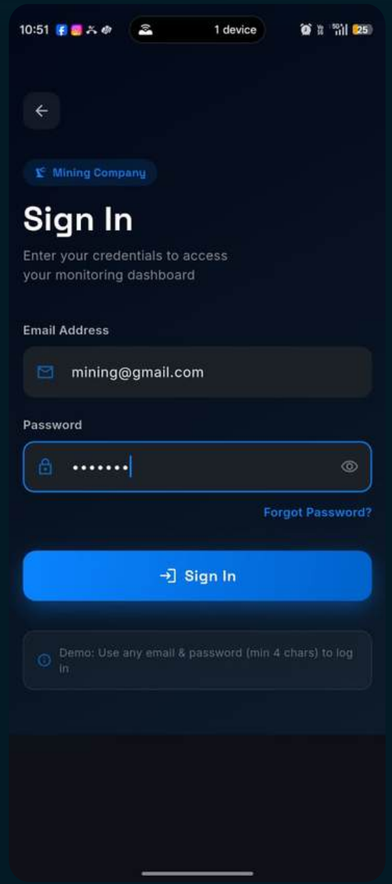


Role-Based Entry

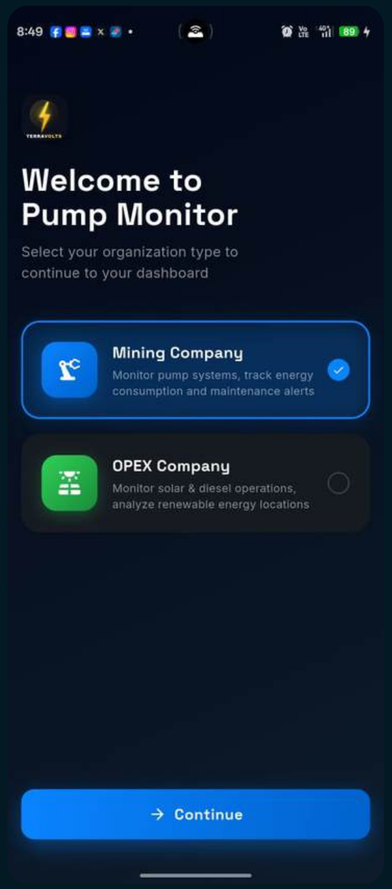
Users first select their specific industry—Mining for pump telemetry and energy alerts, or OPEX for solar-diesel hybrid optimization and site analysis

Secure Authentication

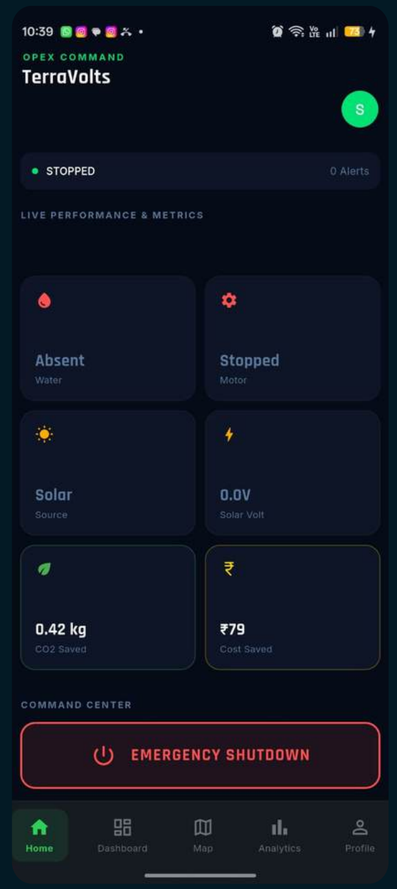
Following selection, a tailored Sign-In portal ensures that organizations can securely access their specific real-time dashboards and energy analytics using verified credentials.



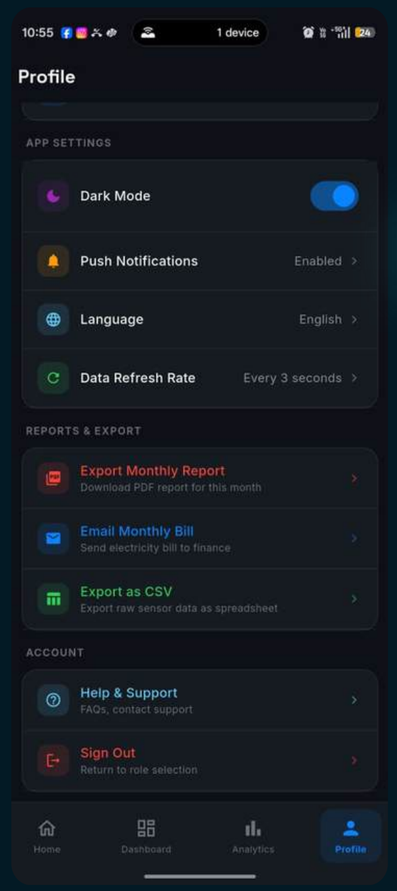
Home and Dashboard



- Real-time monitoring of motor status, solar input, water level, and performance metrics in a single smart dashboard.
- Tracks energy savings, CO_2 reduction, and operational cost efficiency for sustainable operation.
- Provides remote control with emergency shutdown and live alerts for safe and reliable system management.
- Instant Emergency Shutdown feature ensures immediate system stop during critical faults for maximum safety.



SIGN OUT



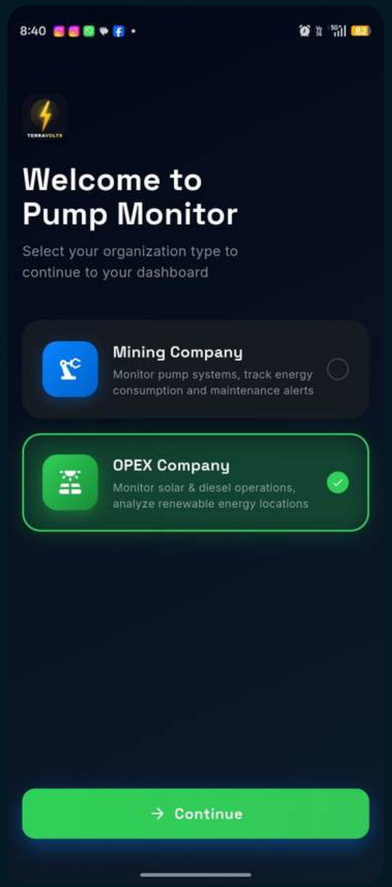
Data Portability

Features professional tools to Export PDF/CSV reports and email billing data, making it easy to transition from live monitoring to long-term documentation.

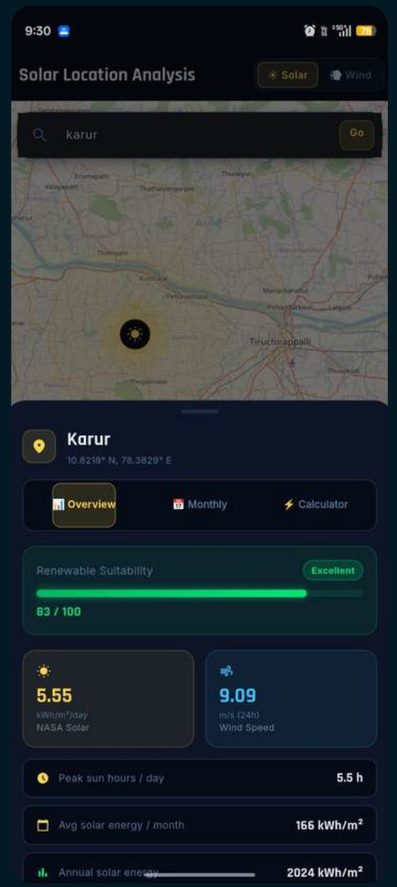
User Security

Provides quick access to Help & Support and a secure Sign Out to manage multi-user or multi-organization access.

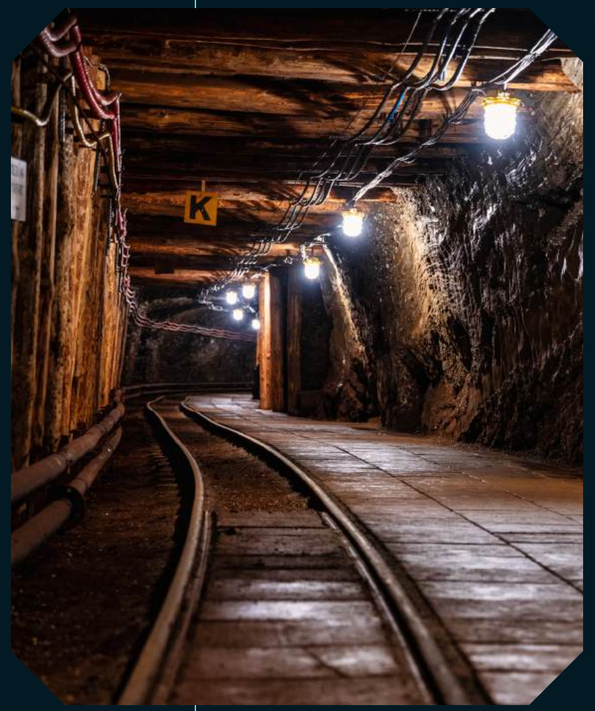
OPEX LOGIN



This Location Energy Insights panel converts geospatial data into a feasibility score (e.g., 83/100) to classify a region's renewable potential as "Excellent." It provides critical metrics like NASA-sourced solar irradiance and wind speeds, alongside actionable projections for peak sun hours and annual energy yield to streamline site selection and project planning.



OUR SERVICES



- Smart App Monitoring
- Intelligent Sun Tracking
- Automatic Power Switching
- Real- Time Water Control
- Smart alerts & Safety

SLIDE – COST ANALYSIS (REALISTIC & JUSTIFIED)

- The total prototype cost is ₹2000, based on actual purchase bills from local suppliers.
- All components are sourced at market-standard prices, ensuring affordability and scalability.
- In mining operations, a typical diesel pump (20–50 HP) consumes around 5–6 liters of diesel per hour.
- With diesel cost ≈ ₹95/L, the operating cost becomes:
- ₹500–₹600 per hour → ₹5,000+ per day → ₹15–20 lakh per year.
- Our system uses solar as the primary source and diesel only as backup, reducing fuel usage by up to 70–80%.
- This reduces annual operating cost to approximately ₹3–5 lakh, saving up to ₹15 lakh per year.
- Strong Closing Line (very important)
- “Our system shifts from fuel-dependent operation to solar-first operation, drastically reducing long-term costs.”

ON SITE RENEWABLE SYSTEM

- Power is generated directly at the mining site (solar/wind), reducing dependency on grid.
- Energy is supplied through a local distribution system with minimal transmission loss.
- Used to operate dewatering pumps, conveyors, and lighting systems.
- Often combined with battery or diesel backup for continuous 24/7 operation.
- Provides cost savings, energy independence, and reduced environmental impact.

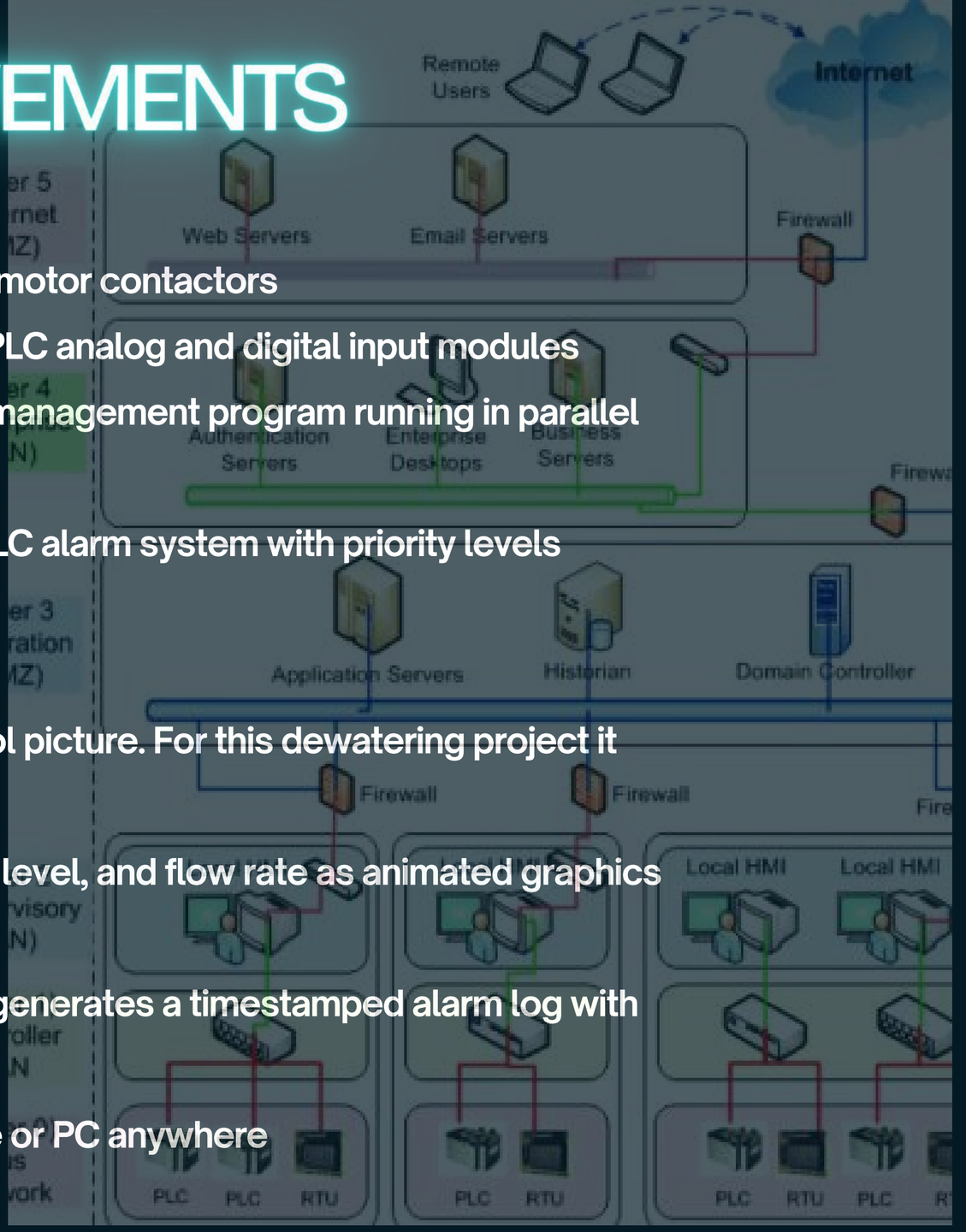
FUTURE SCOPE OF IMPROVEMENTS

PLC Integration

- The relay module gets replaced by PLC digital output modules driving motor contactors
- All sensors (water level, flow S201, voltage, current) wire directly into PLC analog and digital input modules
- The buck/boost and charging logic becomes a dedicated PLC power management program running in parallel with pump control
- Fault detection (dry run, overcurrent, low battery) becomes a proper PLC alarm system with priority levels

SCADA Layer on Top

- SCADA sits above the PLC and gives you the full monitoring and control picture. For this dewatering project it would provide:
- A live mimics screen showing your flooded zone, pump status, battery level, and flow rate as animated graphics — exactly like the simulation we built but as a real industrial HMI
- Alarm management — every fault (dry run, low battery, sensor failure) generates a timestamped alarm log with acknowledgement
- Remote access via OPC-UA or MQTT so you can monitor from a phone or PC anywhere



WE STRIVE TOWARDS A CLEANER AND GREENER ENERGY!!

The future is green energy, sustainability renewable energy